

# **Implementation of pfSense Firewall with Wazuh SIEM for Network Security Monitoring**

## **Objective**

The objective of this task is to deploy and configure pfSense as a software-based firewall in a virtual lab environment and integrate it with Wazuh for centralized log collection and security monitoring. The firewall will be configured with security rules to restrict traffic from selected countries, block access to specified websites, and enforce administrative access controls. The generated firewall events and network activity will be monitored through the Wazuh dashboard to analyze traffic, verify security policies, and understand how firewall logs can support threat detection and Security Operations Center (SOC) activities.

## **Lab Environment**

**Virtualization Platform:** Oracle VirtualBox

**Firewall:** pfSense (Community Edition)

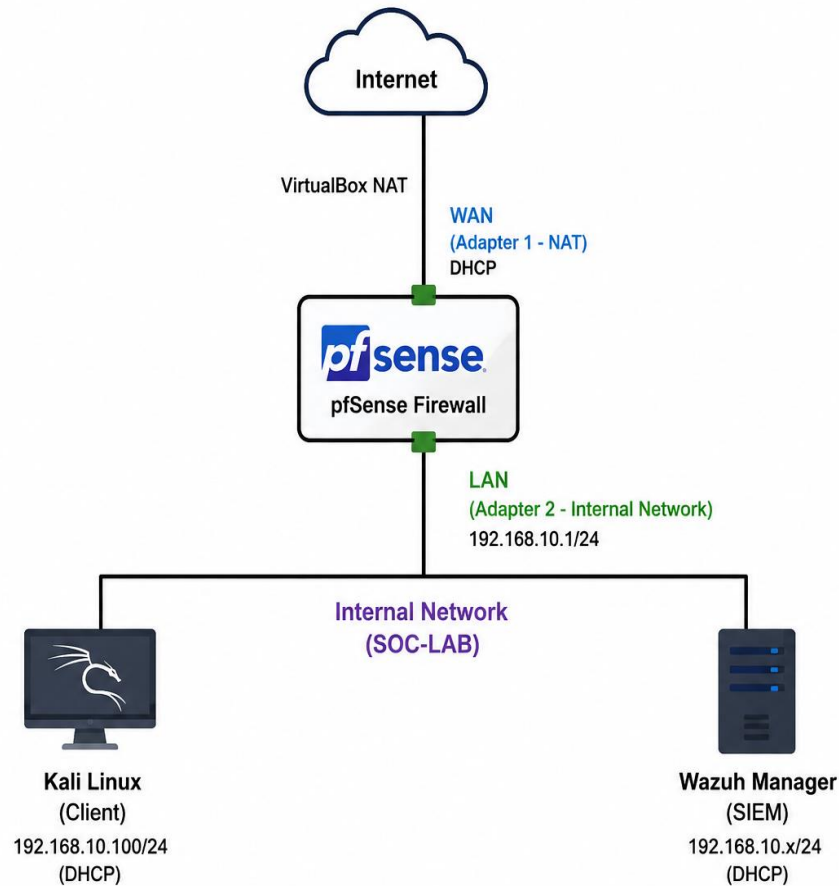
**SIEM Platform:** Wazuh Manager

**Test OS:** Kali Linux

## **Network Structure:**

The lab is designed using Oracle VirtualBox to simulate a smaller enterprise environment. PfSense was deployed as the central firewall and gateway between the external and internal lab network. The pfsense virtual machine was configured with two network interfaces. The WAN was connected to VirtualBox NAT network to provide internet access, while the LAN interface was connected to an internal Network named SOC-LAB. The Kali Linux was connected to SOC-LAB internal network and obtained its IP from pfsense DHCP server. The Wazuh Manager virtual machine is also connected to the internal network to receive and analyze firewall logs generated by pfSense.

This topology ensures that all traffic generated by the client system passes through the pfSense firewall before reaching external networks, allowing firewall rules, traffic filtering, and security monitoring to be applied effectively.



## IP Addressing Table

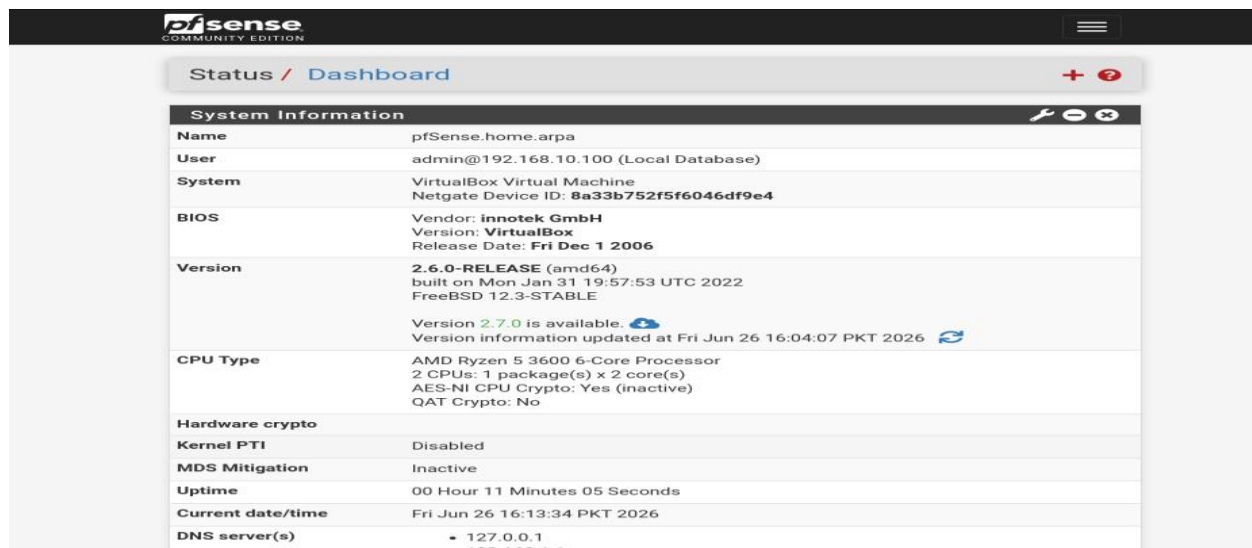
| Device     | Interface | IP Address       | Purpose               |
|------------|-----------|------------------|-----------------------|
| pfSense    | WAN       | 10.0.2.15 (DHCP) | Internet Connectivity |
| pfSense    | LAN       | 192.168.10.1/24  | Gateway & DHCP        |
| Kali Linux | eth0      | 192.168.10.100   | Test Client           |
| Wazuh      | eth0      | 192.168.10.x     | SIEM & Log Analysis   |

## Accessing the pfSense Web Interface

After completing the network configuration, the pfSense WebGUI was accessed from Kali Linux using LAN IP address of the firewall

(<https://192.168.10.1>).

The login credentials are used to login and the initial setup wizard was then completed by configuring hostname, DNS servers, timezone, WAN settings, and administrator password. Once the setup completed, the pfSense dashboard become accessible for further firewall configuration and management.



## Integrating pfSense with Wazuh

The integration of pfSense with Wazuh was performed to enable centralized collection and monitoring of firewall logs within the SIEM platform. This integration allows firewall events generated by pfSense to be forwarded to the wazuh manager, where they can be stored, analyzed, and used for security monitoring and incident investigation.

Wazuh manager was first configured to receive syslog messages by modifying the ossec.conf configuration file. A remote syslog listener was enabled using the UDP protocol on port 514, with the pfSense LAN IP address (192.168.10.1) specified as the allowed source and the wazuh server ip address (192.168.10.101) configured as the

local listening address. After saving the configuration, the wazuh manager service was restarted to apply the changes.

```
GNU nano 8.3 /var/ossec/etc/ossec.conf
<!--
Wazuh - Manager - Default configuration for amzn 2023
More info at: https://documentation.wazuh.com
Mailing list: https://groups.google.com/forum/#!forum/wazuh
-->

<ossec_config>
  <remote>
    <connection>syslog</connection>
    <port>514</port>
    <protocol>udp</protocol>
    <allowed-ips>192.168.10.1</allowed-ips>
    <local-ip>192.168.10.101</local-ip>
  </remote>
</global>
```

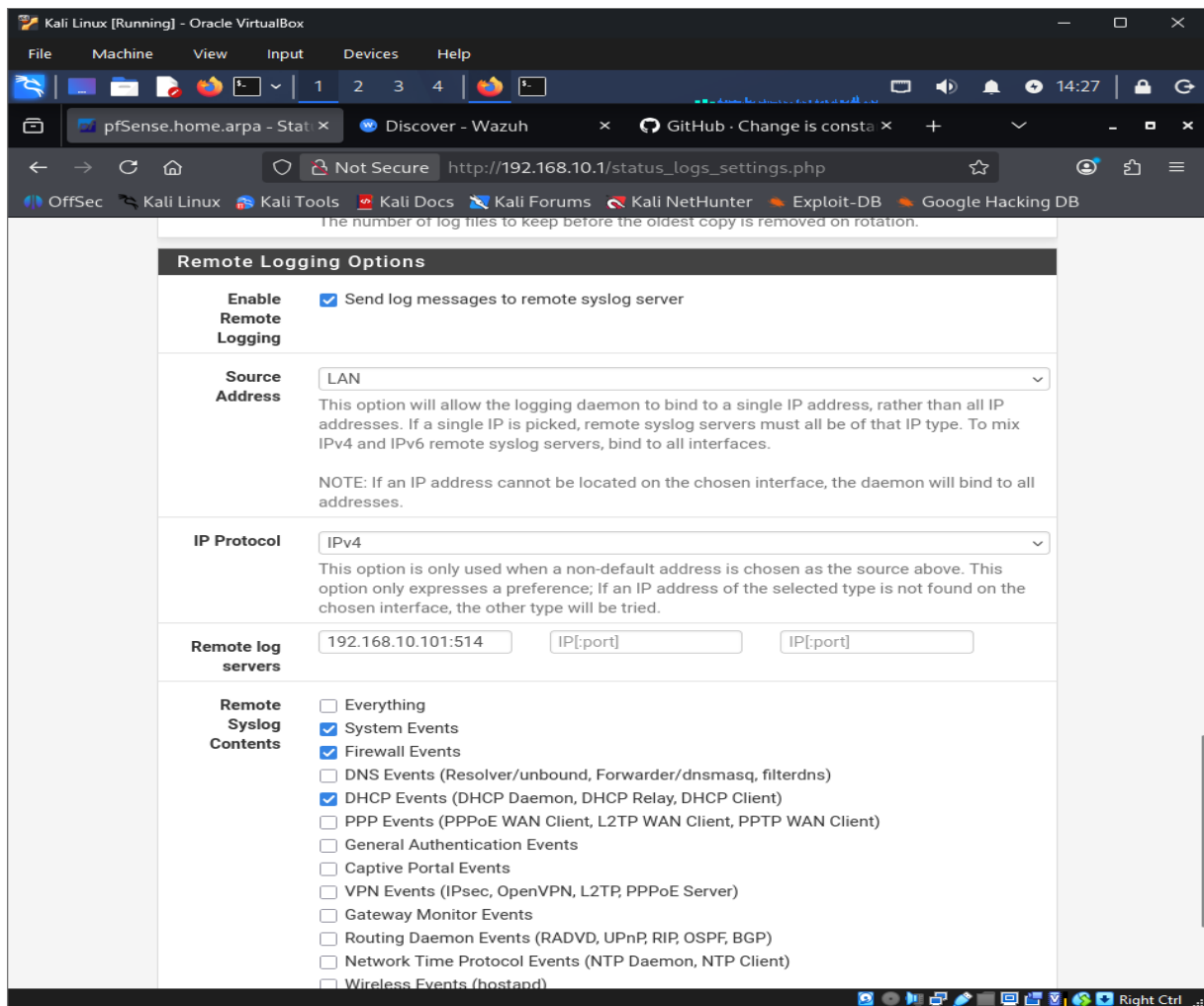
To verify that the wazuh Manager was successfully listening for incoming syslog messages, the UDP listening ports were checked using the ss command. The output confirmed that the wazuh was actively listening on UDP port 514, indicating that it was ready to receive logs from pfSense.

```
[wazuh-user@wazuh-server ~]$ sudo ss -ltn | grep 514
UNCONN 0      0      192.168.10.101:514      0.0.0.0:*
[wazuh-user@wazuh-server ~]$
```

Next, remote logging was configured on pfSense through Status ->

System logs -> Settings. The Wazuh Manager (192.168.10.101) was configured as the remote syslog server using UDP port 514, while the LAN interface was selected as the source address. Firewall Events, System Events, and DHCP Events were enabled to ensure

that security-related logs were forwarded to the SIEM platform.



After completing the configuration, network traffic was generated from kali linux to produce firewall events. The tcpdump utility was executed on the wazuh manager to verify that syslog packet were being received from pfsense. The captured packets contained filterlog entries, confirming successful transmission of firewall logs

over the network.

```
(root@kali)-[/home/user]
# ping -c 10 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=62 time=52.1 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=62 time=45.6 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=62 time=50.4 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=62 time=51.3 ms
64 bytes from 8.8.8.8: icmp_seq=5 ttl=62 time=47.9 ms
64 bytes from 8.8.8.8: icmp_seq=6 ttl=62 time=51.9 ms
64 bytes from 8.8.8.8: icmp_seq=7 ttl=62 time=56.5 ms
64 bytes from 8.8.8.8: icmp_seq=8 ttl=62 time=54.8 ms
64 bytes from 8.8.8.8: icmp_seq=9 ttl=62 time=56.5 ms
64 bytes from 8.8.8.8: icmp_seq=10 ttl=62 time=53.3 ms

— 8.8.8.8 ping statistics —
10 packets transmitted, 10 received, 0% packet loss, time 9064ms
rtt min/avg/max/mdev = 45.646/52.042/56.530/3.334 ms
```

```
[wazuh-user@wazuh-server ~]$ sudo tcpdump -A -i any udp port 514
tcpdump: data link type LINUX_SLL2
dropped privs to tcpdump
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on any, link-type LINUX_SLL2 (Linux cooked v2), snapshot length 262144 bytes
09:58:38.759199 eth0 In IP pfSense.home.arpa.syslog > wazuh-server.syslog: SYSLOG daemon.info, len
gth: 50
E..N...@...
...
e.....:7V<30>Jun 28 14:58:38 dhcp6c[11081]: Sending Solicit
09:58:38.859802 eth0 In IP pfSense.home.arpa.syslog > wazuh-server.syslog: SYSLOG local0.info, len
gth: 143
E...8...@.....
...
e.....
.<134>Jun 28 14:58:38 filterlog[33281]: 83,,,100000101,em1,match,pass,in,4,0x0,,127,23342,0,DF,17,ud
p,73,192.168.10.101,192.168.10.1,59873,53,53
09:58:38.859880 eth0 In IP pfSense.home.arpa.syslog > wazuh-server.syslog: SYSLOG local0.info, len
gth: 143
E...k$.@.yg..
...
e.....<134>Jun 28 14:58:38 filterlog[33281]: 83,,,100000101,em1,match,pass,in,4,0x0,,127,45750,0,
DF,17,udp,71,192.168.10.101,192.168.10.1,49940,53,51
09:58:41.859190 eth0 In IP pfSense.home.arpa.syslog > wazuh-server.syslog: SYSLOG local0.info, len
gth: 143
E...4...@.....
...
e.....A<134>Jun 28 14:58:41 filterlog[33281]: 83,,,100000101,em1,match,pass,in,4,0x0,,64,18430,0,D
F,1,icmp,84,192.168.10.100,8.8.8.8,request,33493,164
```

Finally, the pfSense log decoder was verified using the wazuh-logtest utility. A sample pfSense firewall log was processed through the decoder, which successfully matched the pf decoder and triggered RULE ID 87700. This confirmed that the decoder was

functioning correctly and capable of interpreting pfSense firewall logs.



```
Wazuh v4.14.5 OVA [Running] - Oracle VirtualBox
[wazuh-user@wazuh-server ~]# sudo grep -n "<program_name>" /var/ossec/ruleset/decoders/0455-pfsense_
decoders.xml
18: <program_name>filterlog</program_name>
[wazuh-user@wazuh-server ~]# sudo /var/ossec/bin/wazuh-logtest
Starting wazuh-logtest v4.14.5
Type one log per line

Jun 28 14:02:33 filterlog: 83,,,100000101,em1,match,pass,in,4,0xb0,,64,59586,0,DF,6,TCP,60,192.168.1
0.100,140.248.129.91,41670,443,0,S,2482822653,,64240,,ms;sackOK;TS:nop;wscale

**Phase 1: Completed pre-decoding.
  full event: 'Jun 28 14:02:33 filterlog: 83,,,100000101,em1,match,pass,in,4,0xb0,,64,59586,0,
DF,6,TCP,60,192.168.10.100,140.248.129.91,41670,443,0,S,2482822653,,64240,,ms;sackOK;TS:nop;wscale'
  timestamp: 'Jun 28 14:02:33'
  hostname: 'filterlog:'
  program_name: 'filterlog'

**Phase 2: Completed decoding.
  name: 'pf'
  action: 'pass'
  dstip: '140.248.129.91'
  dstport: '443'
  id: '100000101'
  length: '0'
  protocol: 'TCP'
  srcip: '192.168.10.100'
  srcport: '41670'

**Phase 3: Completed filtering (rules).
  id: '87700'
  level: '0'
  description: 'pfSense firewall rules grouped.'
  groups: '['pfsense']'
  firedtimes: '1'
  mail: 'False'
```

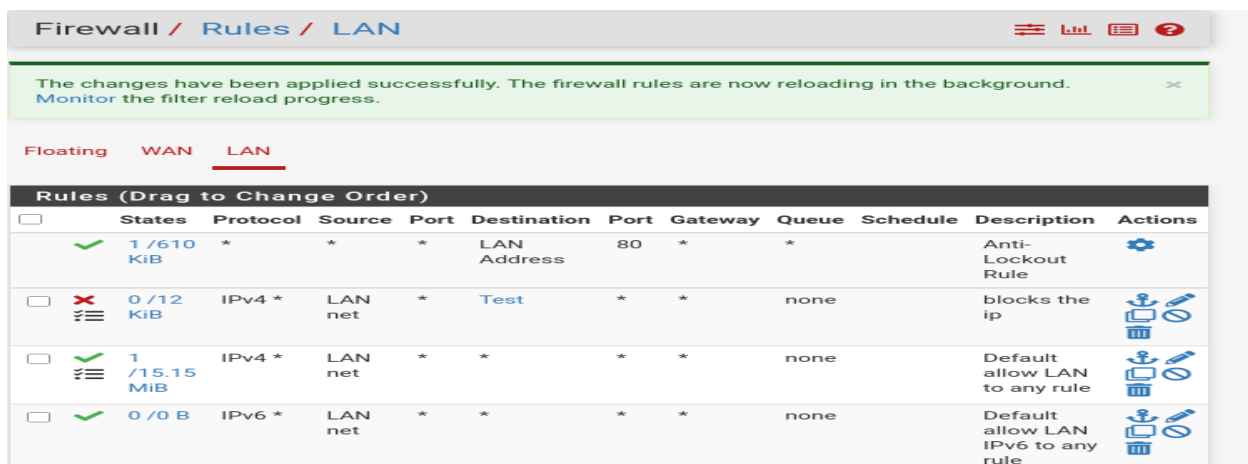
The integration of pfsense with wazuh is successfully established. The wazuh manager was configured to receive remote syslog messages, and connectivity was verified by confirming that UDP



port 514 was actively listening. Firewall logs generated by pfSense were successfully transmitted, received, and archived on the wazuh server. Decoder testing further confirmed that pfsense firewall logs could be correctly interpreted by wazuh, validating the successful integration between the firewall and SIEM Platform.

## Blocking Network Traffic Using pfSense Firewall Rules

A new firewall rule was created under the LAN interface with the action set to Block. The source was configured as LAN net, while the destination is set to a selected IP address (111.68.99.168). The rule was positioned above the default Allow LAN to any rule to ensure that the blocking rule was evaluated before any general allow rule. Logging was enabled on the rule to record all matching traffic in the pfsense firewall logs.



|                                                                                                                                                         |               |          |         |      |             |      |         |       |          |                                    |         |
|---------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|----------|---------|------|-------------|------|---------|-------|----------|------------------------------------|---------|
| Firewall / Rules / LAN                                                                                                                                  |               |          |         |      |             |      |         |       |          |                                    |         |
| The changes have been applied successfully. The firewall rules are now reloading in the background. <a href="#">Monitor the filter reload progress.</a> |               |          |         |      |             |      |         |       |          |                                    |         |
| Floating WAN LAN                                                                                                                                        |               |          |         |      |             |      |         |       |          |                                    |         |
| Rules (Drag to Change Order)                                                                                                                            |               |          |         |      |             |      |         |       |          |                                    |         |
| <input type="checkbox"/>                                                                                                                                | States        | Protocol | Source  | Port | Destination | Port | Gateway | Queue | Schedule | Description                        | Actions |
| <input checked="" type="checkbox"/>                                                                                                                     | 1 / 610 KiB   | *        | *       | *    | LAN Address | 80   | *       | *     |          | Anti-Lockout Rule                  |         |
| <input type="checkbox"/>                                                                                                                                | 0 / 12 KiB    | IPv4 *   | LAN net | *    | Test        | *    | *       | none  |          | blocks the ip                      |         |
| <input type="checkbox"/>                                                                                                                                | 1 / 15.15 MiB | IPv4 *   | LAN net | *    | *           | *    | *       | none  |          | Default allow LAN to any rule      |         |
| <input type="checkbox"/>                                                                                                                                | 0 / 0 B       | IPv6 *   | LAN net | *    | *           | *    | *       | none  |          | Default allow LAN IPv6 to any rule |         |

After applying the firewall rule, connectivity test was performed from the kali linux. Attempts to access the blocked ip address were

unsuccessful, confirming that the firewall rule was functioning.

|   |                 | LAN to any rule<br>(100000101) |                                                 |                      |                          |
|---|-----------------|--------------------------------|-------------------------------------------------|----------------------|--------------------------|
| ✓ | Jun 28 15:55:52 | LAN                            | Default allow<br>LAN to any rule<br>(100000101) | 192.168.10.100:36438 | 203.80.128.20:123 UDP    |
| ✓ | Jun 28 15:56:36 | LAN                            | Default allow<br>LAN to any rule<br>(100000101) | 192.168.10.100:50990 | 192.168.10.1:53 UDP      |
| ✓ | Jun 28 15:56:36 | LAN                            | Default allow<br>LAN to any rule<br>(100000101) | 192.168.10.100:44382 | 151.101.141.91:443 TCP:S |
| ✓ | Jun 28 15:56:36 | LAN                            | Default allow<br>LAN to any rule<br>(100000101) | 192.168.10.100:44384 | 151.101.141.91:443 TCP:S |
| ✓ | Jun 28 15:57:05 | LAN                            | Default allow<br>LAN to any rule<br>(100000101) | 192.168.10.100:58034 | 192.168.10.1:53 UDP      |
| ✓ | Jun 28 15:57:05 | LAN                            | Default allow<br>LAN to any rule<br>(100000101) | 192.168.10.100:42484 | 34.107.243.93:443 TCP:S  |
| ✓ | Jun 28 15:57:05 | LAN                            | Default allow<br>LAN to any rule<br>(100000101) | 192.168.10.100:42498 | 34.107.243.93:443 TCP:S  |
| ✓ | Jun 28 15:57:06 | LAN                            | Default allow<br>LAN to any rule<br>(100000101) | 192.168.10.100:42512 | 34.107.243.93:443 TCP:S  |
| ✗ | Jun 28 16:03:52 | LAN                            | blocks the ip<br>(1782642808)                   | 192.168.10.100       | 111.68.99.168 ICMP       |
| ✗ | Jun 28 16:03:53 | LAN                            | blocks the ip<br>(1782642808)                   | 192.168.10.100       | 111.68.99.168 ICMP       |
| ✗ | Jun 28 16:03:54 | LAN                            | blocks the ip<br>(1782642808)                   | 192.168.10.100       | 111.68.99.168 ICMP       |
| ✗ | Jun 28 16:03:55 | LAN                            | blocks the ip<br>(1782642808)                   | 192.168.10.100       | 111.68.99.168 ICMP       |
| ✗ | Jun 28 16:03:56 | LAN                            | blocks the ip<br>(1782642808)                   | 192.168.10.100       | 111.68.99.168 ICMP       |
| ✗ | Jun 28 16:03:57 | LAN                            | blocks the ip<br>(1782642808)                   | 192.168.10.100       | 111.68.99.168 ICMP       |
| ✗ | Jun 28 16:03:58 | LAN                            | blocks the ip<br>(1782642808)                   | 192.168.10.100       | 111.68.99.168 ICMP       |
| ✗ | Jun 28 16:03:59 | LAN                            | blocks the ip<br>(1782642808)                   | 192.168.10.100       | 111.68.99.168 ICMP       |
| ✗ | Jun 28 16:04:00 | LAN                            | blocks the ip<br>(1782642808)                   | 192.168.10.100       | 111.68.99.168 ICMP       |
| ✗ | Jun 28 16:04:01 | LAN                            | blocks the ip<br>(1782642808)                   | 192.168.10.100       | 111.68.99.168 ICMP       |

The blocked firewall events were also forwarded to the wazuh manager the previously configured syslog integration. The archived logs on the wazuh server contained the corresponding filterlog entries which confirm that firewall events continued to be collected and archived for centralized monitoring and analysis.

```
Wazuh v4.14.5 OVA [Running] - Oracle VirtualBox
2026 Jun 28 11:03:53 wazuh-server->192.168.10.1 Jun 28 16:03:53 filterlog[33873]: 83,,,1782642808,en
1,match,block,in,4,0x0,,64,9621,0,DF,1,icmp,84,192.168.10.100,111.68.99.168,request,51314,264
2026 Jun 28 11:03:54 wazuh-server->192.168.10.1 Jun 28 16:03:54 filterlog[33873]: 83,,,1782642808,en
1,match,block,in,4,0x0,,64,9640,0,DF,1,icmp,84,192.168.10.100,111.68.99.168,request,51314,364
2026 Jun 28 11:03:55 wazuh-server->192.168.10.1 Jun 28 16:03:55 filterlog[33873]: 83,,,1782642808,en
1,match,block,in,4,0x0,,64,9880,0,DF,1,icmp,84,192.168.10.100,111.68.99.168,request,51314,464
2026 Jun 28 11:03:56 wazuh-server->192.168.10.1 Jun 28 16:03:56 filterlog[33873]: 83,,,1782642808,en
1,match,block,in,4,0x0,,64,10023,0,DF,1,icmp,84,192.168.10.100,111.68.99.168,request,51314,564
2026 Jun 28 11:03:57 wazuh-server->192.168.10.1 Jun 28 16:03:57 filterlog[33873]: 83,,,1782642808,en
1,match,block,in,4,0x0,,64,10113,0,DF,1,icmp,84,192.168.10.100,111.68.99.168,request,51314,664
2026 Jun 28 11:03:58 wazuh-server->192.168.10.1 Jun 28 16:03:58 filterlog[33873]: 83,,,1782642808,en
1,match,block,in,4,0x0,,64,10164,0,DF,1,icmp,84,192.168.10.100,111.68.99.168,request,51314,764
2026 Jun 28 11:03:59 wazuh-server->192.168.10.1 Jun 28 16:03:59 filterlog[33873]: 83,,,1782642808,en
1,match,block,in,4,0x0,,64,10308,0,DF,1,icmp,84,192.168.10.100,111.68.99.168,request,51314,864
2026 Jun 28 11:04:00 wazuh-server->192.168.10.1 Jun 28 16:04:00 filterlog[33873]: 83,,,1782642808,en
1,match,block,in,4,0x0,,64,10437,0,DF,1,icmp,84,192.168.10.100,111.68.99.168,request,51314,964
2026 Jun 28 11:04:01 wazuh-server->192.168.10.1 Jun 28 16:04:01 filterlog[33873]: 83,,,1782642808,en
1,match,block,in,4,0x0,,64,10578,0,DF,1,icmp,84,192.168.10.100,111.68.99.168,request,51314,1064
2026 Jun 28 11:12:56 wazuh-server->192.168.10.1 Jun 28 16:12:56 filterlog[33873]: 84,,,100000101,en1
,match,pass,in,4,0xb8,,64,22948,0,DF,17,udp,76,192.168.10.100,203.80.128.20,41752,123,56
2026 Jun 28 11:13:51 wazuh-server->192.168.10.1 Jun 28 16:13:51 filterlog[33873]: 84,,,100000101,en1
,match,pass,in,4,0x0,,64,2942,0,DF,17,udp,73,192.168.10.100,192.168.10.1,39238,53,53
2026 Jun 28 11:13:51 wazuh-server->192.168.10.1 Jun 28 16:13:51 filterlog[33873]: 84,,,100000101,en1
,match,pass,in,4,0x0,,64,61259,0,DF,6,tcp,60,192.168.10.100,142.251.38.42,50890,443,0,S,2810073982,,
64240,,mss:sackOK;TS:nop:wscale
2026 Jun 28 11:17:11 wazuh-server->192.168.10.1 Jun 28 16:17:11 filterlog[33873]: 84,,,100000101,en1
,match,pass,in,4,0x0,,64,33034,0,DF,17,udp,71,192.168.10.100,192.168.10.1,58744,53,51
2026 Jun 28 11:17:11 wazuh-server->192.168.10.1 Jun 28 16:17:11 filterlog[33873]: 84,,,100000101,en1
,match,pass,in,4,0x0,,64,34443,0,DF,6,tcp,60,192.168.10.100,34.107.243.93,47434,443,0,S,1366632144,,
64240,,mss:sackOK;TS:nop:wscale
2026 Jun 28 11:17:12 wazuh-server->192.168.10.1 Jun 28 16:17:12 filterlog[33873]: 84,,,100000101,en1
,match,pass,in,4,0x0,,64,36949,0,DF,6,tcp,60,192.168.10.100,34.107.243.93,47442,443,0,S,1700884666,,
64240,,mss:sackOK;TS:nop:wscale
2026 Jun 28 11:17:12 wazuh-server->192.168.10.1 Jun 28 16:17:12 filterlog[33873]: 84,,,100000101,en1
,match,pass,in,4,0x0,,64,48427,0,DF,6,tcp,60,192.168.10.100,34.107.243.93,49024,443,0,S,481894964,,6
4240,,mss:sackOK;TS:nop:wscale
[wazuh-user@wazuh-server ~]$_
```

The firewall rule successfully prevented the kali linux from accessing the specified destination ip address. The blocked traffic was recorded in the pfSense Firewall Logs and forwarded to the Wazuh Manager through Syslog successfully.

## Implementing Administrative Access Controls

To restrict administrative access to the pfSense firewall, firewall rules were configured to allow only kali linux machine (192.168.10.100) to access WebGUI of pfsense. An allow rule was created for kali linux with destination set to This Firewall. A block rule was then created below to deny access from all other devices on the LAN. Logging was enabled on the block rule to record the unauthorized access attempts.

Since pfsense automatically allows WebGUI access through the Anti-lockout rule, that rule was disabled so that the user-defined firewall rules can control administrative access.

Firewall / Rules / LAN    

Floating WAN LAN

| Rules (Drag to Change Order) |                                                                                                     |          |                |      |               |      |         |       |          |                                    |                                                                                                                                                                                                                                                                                                                                                         |
|------------------------------|-----------------------------------------------------------------------------------------------------|----------|----------------|------|---------------|------|---------|-------|----------|------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <input type="checkbox"/>     | States                                                                                              | Protocol | Source         | Port | Destination   | Port | Gateway | Queue | Schedule | Description                        | Actions                                                                                                                                                                                                                                                                                                                                                 |
| <input type="checkbox"/>     |  2 / 58 KiB      | IPv4 TCP | 192.168.10.100 | *    | This Firewall | *    | *       | none  |          | Allow pfSense WebGUI from Kali     |     |
| <input type="checkbox"/>     |  0 / 2 KiB       | IPv4 TCP | LAN net        | *    | This Firewall | *    | *       | none  |          | Block WebGUI from all other hosts  |     |
| <input type="checkbox"/>     |  0 / 0 B         | IPv4 *   | LAN net        | *    | Test          | *    | *       | none  |          | blocks the ip                      |     |
| <input type="checkbox"/>     |  45 / 286.98 MiB | IPv4 *   | LAN net        | *    | *             | *    | *       | none  |          | Default allow LAN to any rule      |     |
| <input type="checkbox"/>     |  0 / 0 B         | IPv6 *   | LAN net        | *    | *             | *    | *       | none  |          | Default allow LAN IPv6 to any rule |     |

 Add  Add  Delete  Save  Separator

**pfSense** COMMUNITY EDITION System ▾ Interfaces ▾ Firewall ▾ Services ▾ VPN ▾ Status ▾ Diagnostics ▾ Help ▾

System / [Advanced](#) / [Admin Access](#) ?

[Admin Access](#) [Firewall & NAT](#) [Networking](#) [Miscellaneous](#) [System Tunables](#) [Notifications](#)

### webConfigurator

|                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Protocol</b>                  | <input checked="" type="radio"/> HTTP <input type="radio"/> HTTPS (SSL/TLS)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>TCP port</b>                  | <input type="text" value="80"/> <p>Enter a custom port number for the webConfigurator above to override the default (80 for HTTP, 443 for HTTPS). Changes will take effect immediately after save.</p>                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Max Processes</b>             | <input type="text" value="2"/> <p>Enter the number of webConfigurator processes to run. This defaults to 2. Increasing this will allow more users/browsers to access the GUI concurrently.</p>                                                                                                                                                                                                                                                                                                                                                                              |
| <b>WebGUI redirect</b>           | <input type="checkbox"/> Disable webConfigurator redirect rule<br>When this is unchecked, access to the webConfigurator is always permitted even on port 80, regardless of the listening port configured. Check this box to disable this automatically added redirect rule.                                                                                                                                                                                                                                                                                                 |
| <b>OCSP Must-Staple</b>          | <input type="checkbox"/> Force OCSP Stapling in nginx<br>When this is checked, OCSP Stapling is forced on in nginx. Remember to upload your certificate as a full chain, not just the certificate, or this option will be ignored by nginx.                                                                                                                                                                                                                                                                                                                                 |
| <b>WebGUI Login Autocomplete</b> | <input checked="" type="checkbox"/> Enable webConfigurator login autocomplete<br>When this is checked, login credentials for the webConfigurator may be saved by the browser. While convenient, some security standards require this to be disabled. Check this box to enable autocomplete on the login form so that browsers will prompt to save credentials (NOTE: Some browsers do not respect this option).                                                                                                                                                             |
| <b>WebGUI login messages</b>     | <input type="checkbox"/> Disable logging of webConfigurator successful logins<br>When this is checked, successful logins to the webConfigurator will not be logged.                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Anti-lockout</b>              | <input checked="" type="checkbox"/> Disable webConfigurator anti-lockout rule<br>When this is unchecked, access to the webConfigurator on the LAN interface is always permitted, regardless of the user-defined firewall rule set. Check this box to disable this automatically added rule, so access to the webConfigurator is controlled by the user-defined firewall rules (ensure a firewall rule is in place that allows access, to avoid being locked out!)<br><i>Hint: the "Set interface(s) IP address" option in the console menu resets this setting as well.</i> |

The configuration was verified by accessing the pfSense WebGUI from both clients, kali linux (192.168.10.100) and windows (192.168.10.102).

The kali linux machine successfully accessed the WebGUI, while the windows machine was denied access. The blocked connection was also recorded in the pfSense Firewall logs.

Kali Linux [Running] - Oracle VM VirtualBox

File Machine View Input Devices Help

Discover - Wazuh x pfSense.home.arpa - Sta

Not Secure http://192.168.10.1

pfSense  
COMMUNITY EDITION

Status / Dashboard

System Information

|                   |                                                                                                                              |
|-------------------|------------------------------------------------------------------------------------------------------------------------------|
| Name              | pfSense.home.arpa                                                                                                            |
| User              | admin@192.168.10.100 (Local Database)                                                                                        |
| System            | VirtualBox Virtual Machine<br>Netgate Device ID: 8a33b752f5f6046d9e4                                                         |
| BIOS              | Vendor: innotek GmbH<br>Version: VirtualBox<br>Release Date: Fri Dec 1 2006                                                  |
| Version           | 2.6.0-RELEASE (amd64)<br>built on Mon Jan 31 19:57:53 UTC 2022<br>FreeBSD 12.3-STABLE<br><br>Obtaining update status         |
| CPU Type          | AMD Ryzen 5 3600 6-Core Processor<br>2 CPUs: 1 package(s) x 2 core(s)<br>AES-NI CPU Crypto: Yes (inactive)<br>QAT Crypto: No |
| Hardware crypto   |                                                                                                                              |
| Kernel PTI        | Disabled                                                                                                                     |
| MDS Mitigation    | Inactive                                                                                                                     |
| Uptime            | 01 Hour 31 Minutes 42 Seconds                                                                                                |
| Current date/time | Sun Jun 28 18:55:02 PKT 2026                                                                                                 |
| DNS server(s)     | • 127.0.0.1<br>• 8.8.8.8<br>• 1.1.1.1                                                                                        |

Windows 10 [Running] - Oracle VM VirtualBox

192.168.10.1

192.168.10.1

Hmm... can't reach this page

192.168.10.1 took too long to respond

Try:

- Checking the connection
- Checking the proxy and the firewall
- Running Windows Network Diagnostics

ERR\_CONNECTION\_TIMED\_OUT

Details

|   |                 |     |                                                |                      |                 |
|---|-----------------|-----|------------------------------------------------|----------------------|-----------------|
| ✓ | Jun 28 18:57:24 | LAN | Default allow LAN to any rule (100000101)      | 192.168.10.102:63681 | 192.168.10.1:53 |
| ✓ | Jun 28 18:57:24 | LAN | Default allow LAN to any rule (100000101)      | 192.168.10.102:64018 | 20.42.73.26:443 |
| ✓ | Jun 28 18:57:24 | LAN | Default allow LAN to any rule (100000101)      | 192.168.10.102:59720 | 192.168.10.1:53 |
| ✗ | Jun 28 18:57:22 | LAN | Block WebGUI from all other hosts (1782646585) | 192.168.10.102:53516 | 192.168.10.1:80 |
| ✗ | Jun 28 18:57:22 | LAN | Block WebGUI from all other hosts (1782646585) | 192.168.10.102:62650 | 192.168.10.1:80 |
| ✗ | Jun 28 18:57:22 | LAN | Block WebGUI from all other hosts (1782646585) | 192.168.10.102:49332 | 192.168.10.1:80 |

## Implementation Country-Based Traffic Blocking

To block traffic from selected countries, the pfBlockerNG-devel is installed and configured on pfSense. A MaxMind Geolite2 account was created and the id and license key is added to pfBlockerNG to download the GeoIP database. The Asia Geoip list was then configured to block traffic from India, China and Russia, then the configuration was applied using the pfBlockerNG update process. This automatically created the required firewall rules (pfBAsiav4 and pfBAsiav6) to enforce the GeoIP policy.


COMMUNITY EDITION

System / Package Manager / Installed Packages

Installed Packages Available Packages

| Name                | Category | Version  | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Actions                                                                                                                                                                                                                                                     |
|---------------------|----------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ✓ pfBlockerNG-devel | net      | 3.2.0_20 | <p>pfBlockerNG-devel is the Next Generation of pfBlockerNG. Manage IPv4/v6 List Sources into 'Deny, Permit or Match' formats. GeoIP database by MaxMind Inc. (GeoLite2 Free version). De-Duplication, Suppression, and Reputation enhancements. Provision to download from diverse List formats. Advanced Integration for Proofpoint ET IQRisk IP Reputation Threat Sources. Domain Name (DNSBL) blocking via Unbound DNS Resolver.</p> <p>Package Dependencies:</p> <div> lighttpd-1.4.72 jq-1.7_1 gnugrep-3.11<br/> rsync-3.4.0 py-maxminddb-2.4.0<br/> libmaxminddb-1.7.1_1 iprange-1.0.4<br/> grepcidr-2.0 python311-3.11.3 php82-8.2.6<br/> php82-intl-8.2.6 py-sqlite3-3.11.6_8 </div> |    |

 = Update
 = Current

 = Remove
 = Information
 = Reinstall

### MaxMind GeoIP configuration

|                                   |                                                                                                                                                                                                                                                                                                                                               |
|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Attribution</b>                | <p>GeoIP database GeoLite2 distributed under the Creative Commons Attribution-ShareAlike 4.0 International License by: <a href="#">MaxMind Inc.</a></p> <p>The GeoIP database is automatically updated each day at a random hour.</p>                                                                                                         |
| <b>MaxMind Account ID</b>         | <input type="text" value="137"/> <p>To utilize the free MaxMind GeoLite2 GeoIP functionality, you must first register for a free MaxMind user account. Visit the following <a href="#">Link to Register</a> for a free MaxMind user account. Use the GeoIP Update version 3.1.1 or newer registration option.</p>                             |
| <b>MaxMind License Key</b>        | <input type="text" value="hT0siu_Og8cL...Sq9hsc...mmk"/> <p>To utilize the free MaxMind GeoLite2 GeoIP functionality, you must first register for a free MaxMind user account. Visit the following <a href="#">Link to Register</a> for a free MaxMind user account. Utilize the GeoIP Update version 3.1.1 or newer registration option.</p> |
| <b>MaxMind Localized Language</b> | <div>English</div> <p>Select the localized name data from the Language options available. Changes to the Locale will be executed in the background, and will take a few minutes to complete. Upon completion, a pfSense Notice will be generated.</p>                                                                                         |

Continent - Asia

Links

Firewall Alias

Firewall Rules

Firewall Logs

NOTES:

GeoIP data by MaxMind Inc. - GeoLite2  
Click here for IMPORTANT info -> **What's new in GeoIP2**

pfSense by default implicitly blocks all unsolicited inbound traffic to the WAN interface.  
Therefore adding GeoIP based firewall rules to the WAN will **not** provide any benefit, unless there are open WAN ports.

It's also **not** recommended to block the 'world', instead consider rules to 'Permit' traffic from selected Countries only.  
Also consider protecting just the specific open WAN ports and it's just as important to protect the outbound LAN traffic.

GeoIP ISOs can also be configured in the pfBlockerNG IPv4/IPv6 Alias(es) Source Definitions (Format: GeoIP)

Use **CTRL+CLICK** to **select/unselect** the IPv4/6 Countries below as required.

AA ASIA UNDEFINED 6255147 (181)

AA ASIA UNDEFINED 6255147\_rep (0)

Afghanistan [1149361] AF (219)

Afghanistan [1149361] AF\_rep (0)

Armenia [174982] AM (333)

Armenia [174982] AM\_rep (82)

Azerbaijan [587116] AZ (396)

Azerbaijan [587116] AZ\_rep (21)

Bahrain [290291] BH (233)

Bahrain [290291] BH\_rep (1)

Bangladesh [1210997] BD (2617)

Bangladesh [1210997] BD\_rep (99)

Bhutan [1252634] BT (128)

Bhutan [1252634] BT\_rep (2)

British Indian Ocean Territory [1282588] IO (39)

British Indian Ocean Territory [1282588] IO\_rep (0)

Brunei [1820814] BN (117)

Brunei [1820814] BN\_rep (0)

Cambodia [1831722] KH (522)

Cambodia [1831722] KH\_rep (8)

**China [1814991] CN (7923)**

**China [1814991] CN\_rep (929)**

Cocos (Keeling) Islands [1547376] CC (17)

Cocos (Keeling) Islands [1547376] CC\_rep (0)

Georgia [614540] GE (488)

Georgia [614540] GE\_rep (110)

Hong Kong [1819730] HK (11343)

Hong Kong [1819730] HK\_rep (3999)

**India [1269750] IN (13021)**

**India [1269750] IN\_rep (1183)**

Indonesia [1643084] ID (6447)

Indonesia [1643084] ID\_rep (131)

Iran [130758] IR (2039)

Iran [130758] IR\_rep (178)

Iraq [99237] IQ (685)

Iraq [99237] IQ\_rep (26)

**Israel [294640] IL (2105)**

**Israel [294640] IL\_rep (597)**

Japan [1861060] JP (12205)

AA ASIA UNDEFINED 6255147 (85)

Afghanistan [1149361] AF (476)

Armenia [174982] AM (432)

Azerbaijan [587116] AZ (242)

Bahrain [290291] BH (190)

Bangladesh [1210997] BD (1900)

Bhutan [1252634] BT (106)

British Indian Ocean Territory [1282588] IO (61)

Brunei [1820814] BN (72)

Cambodia [1831722] KH (499)

**China [1814991] CN (10974)**

Cocos (Keeling) Islands [1547376] CC (23)

Georgia [614540] GE (239)

Hong Kong [1819730] HK (4569)

**India [1269750] IN (7222)**

Indonesia [1643084] ID (4842)

Iran [130758] IR (704)

Iraq [99237] IQ (911)

**Israel [294640] IL (1724)**

Japan [1861060] JP (5883)

Jordan [248816] JO (159)

Kazakhstan [1522867] KZ (971)

Kuwait [285570] KW (290)

Kyrgyzstan [1527747] KG (166)

Laos [1655842] LA (105)

Lebanon [272108] LB (216)

Malaco [1821273] MD (612)

Malaysia [1733045] MY (1332)

Maldives [1282028] MV (1563)

Mongolia [2029969] MN (244)

Myanmar [1327865] MM (330)

Nepal [1282988] NP (265)

North Korea [1873107] KP (537)

Oman [286963] OM (161)

Pakistan [1168579] PK (612)

Palestine [6254930] PS (148)

Philippines [1694008] PH (1030)

Qatar [289088] QA (238)

Saudi Arabia [1102358] SA (1688)

pfSense

COMMUNITY EDITION

Firewall / Rules / WAN

Floating

**WAN**

LAN

Rules (Drag to Change Order)

|                          | States                                 | Protocol | Source                                 | Port | Destination | Port | Gateway | Queue | Schedule | Description                | Actions                                                 |
|--------------------------|----------------------------------------|----------|----------------------------------------|------|-------------|------|---------|-------|----------|----------------------------|---------------------------------------------------------|
| <input type="checkbox"/> | <div><div></div><div>0/0 B</div></div> | *        | Reserved<br>Not<br>assigned<br>by IANA | *    | *           | *    | *       | *     |          | Block<br>bogon<br>networks | <div></div>                                             |
| <input type="checkbox"/> | <div><div></div><div>0/0 B</div></div> | IPv4 *   | pfB_<br>Asia_v4                        | *    | *           | *    | *       | none  |          | pfB_Asia_v4<br>auto rule   | <div><div></div><div></div><div></div><div></div></div> |
| <input type="checkbox"/> | <div><div></div><div>0/0 B</div></div> | IPv6 *   | pfB_<br>Asia_v6                        | *    | *           | *    | *       | none  |          | pfB_Asia_v6<br>auto rule   | <div><div></div><div></div><div></div><div></div></div> |

Add

Add

Delete

Toggle

Copy

Save

Separator



This configuration was verified by attempting to connect to IP addresses located in the selected countries. The firewall successfully blocked the connections, and the blocked traffic was recorded in the pfSense Firewall Logs.

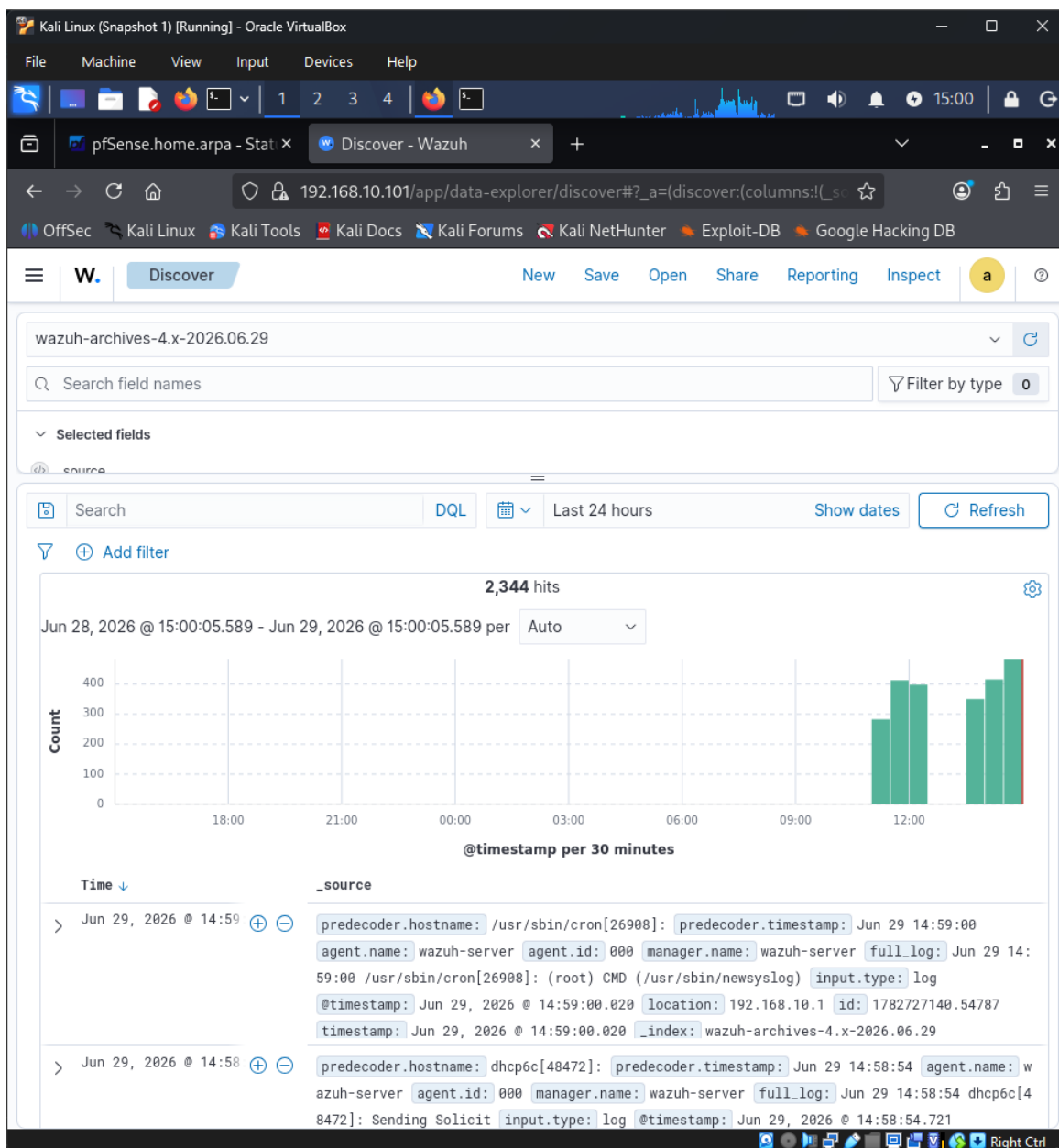
```
(user@kali)-[~]
$ ping -c 3 49.36.0.1
PING 49.36.0.1 (49.36.0.1) 56(84) bytes of data.
^C
— 49.36.0.1 ping statistics —
3 packets transmitted, 0 received, 100% packet loss, time 2027ms

(user@kali)-[~]
$ ping -c 3 1.119.0.0
PING 1.119.0.0 (1.119.0.0) 56(84) bytes of data.
^C
— 1.119.0.0 ping statistics —
3 packets transmitted, 0 received, 100% packet loss, time 2035ms
```

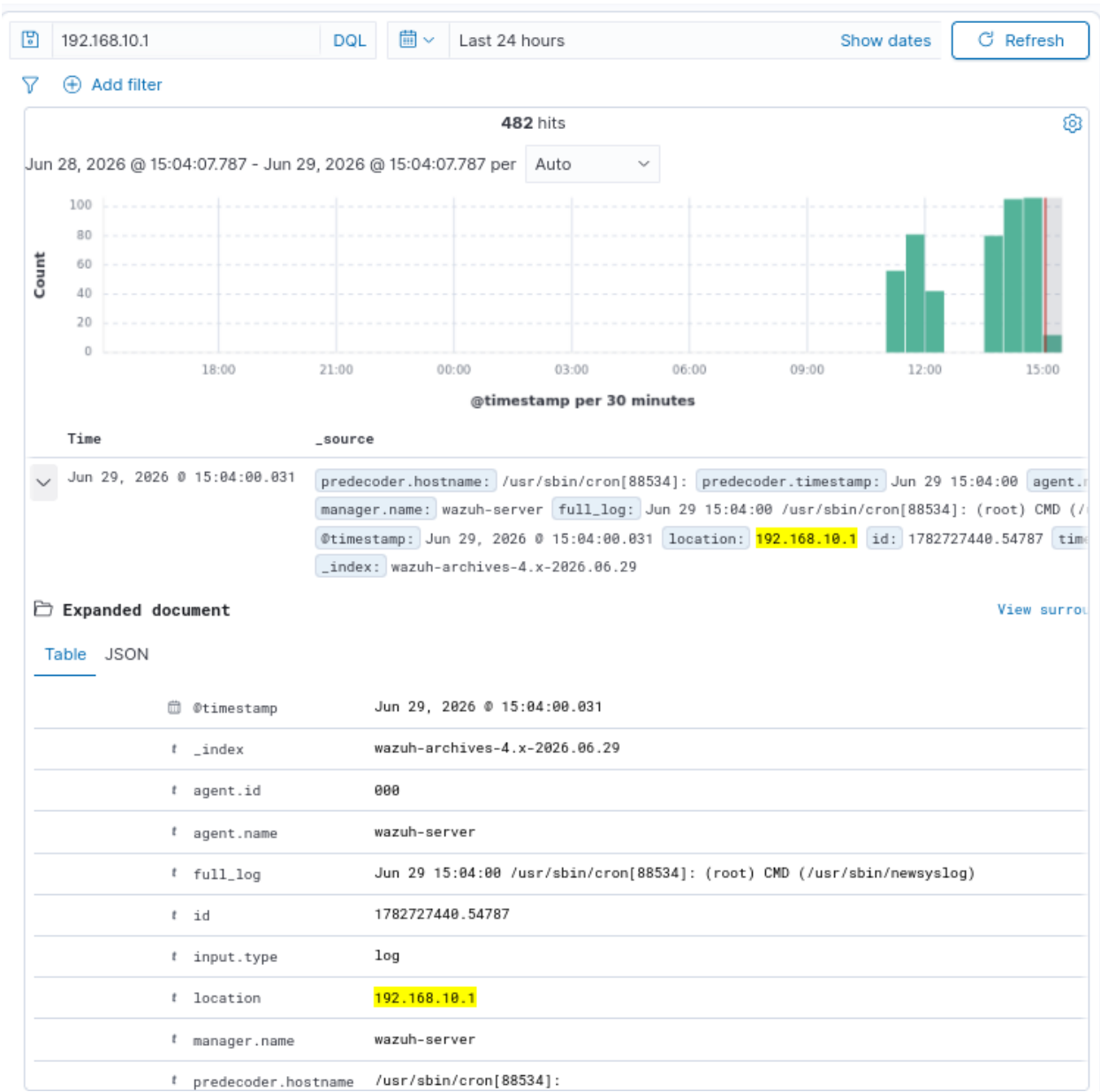
|   |                 |     |                                                                                                                              |                                                                                                    |                                                                                                 |
|---|-----------------|-----|------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| ✗ | Jun 28 23:35:15 | LAN |  pfB_Asia_v4<br>auto rule<br>(1770009341)   |  192.168.10.100   |  49.36.0.1   |
| ✗ | Jun 28 23:35:16 | LAN |  pfB_Asia_v4<br>auto rule<br>(1770009341) |  192.168.10.100 |  49.36.0.1 |
| ✗ | Jun 28 23:35:17 | LAN |  pfB_Asia_v4<br>auto rule<br>(1770009341) |  192.168.10.100 |  49.36.0.1 |
| ✗ | Jun 28 23:35:35 | LAN |  pfB_Asia_v4<br>auto rule<br>(1770009341) |  192.168.10.100 |  1.119.0.0 |
| ✗ | Jun 28 23:35:36 | LAN |  pfB_Asia_v4<br>auto rule<br>(1770009341) |  192.168.10.100 |  1.119.0.0 |
| ✗ | Jun 28 23:35:37 | LAN |  pfB_Asia_v4<br>auto rule<br>(1770009341) |  192.168.10.100 |  1.119.0.0 |

# Monitoring pfSense Logs in Wazuh

After configuring pfSense to forward firewall logs to wazuh server using the syslog protocol, the logs were successfully received and indexed by wazuh. The events were monitored through the Discover page of Wazuh Dashboard, where firewall activities such as allowed and blocked network traffic were displayed in real time.



The received logs were filtered using the pfSense firewall ip address to analyze only the events generated by the firewall. The log enteries contained useful information such as the event timestamp, firewall action, protocol etc. This allowed security events to be monitored and analyzed from a centralized dashboard.



The monitoring process was verified by generating network traffic from kali linux machine. The corresponding firewall events appeared in the Wazuh dashboard shortly after they were generated, confirming that the log forwarding and monitoring configuration was functioning properly.

```
(user@kali)-[~]
$ ping 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=62 time=36.0 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=62 time=36.8 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=62 time=34.5 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=62 time=34.4 ms
64 bytes from 8.8.8.8: icmp_seq=5 ttl=62 time=35.0 ms
^C
--- 8.8.8.8 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4084ms
rtt min/avg/max/mdev = 34.435/35.324/36.759/0.902 ms

(user@kali)-[~]
$ ping 1.2.3.4
PING 1.2.3.4 (1.2.3.4) 56(84) bytes of data.
^C
--- 1.2.3.4 ping statistics ---
5 packets transmitted, 0 received, 100% packet loss, time 4074ms
```

|   |                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|---|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| > | Jun 29, 2026 @ 15:07:00.057 | predecoder.hostname: /usr/sbin/cron[22840]: predecoder.timestamp: Jun 29 15:07:00 agent.name: wazuh-server agent.id: 000 manager.name: wazuh-server full_log: Jun 29 15:07:00 /usr/sbin/cron[22840]: (root) CRD (/usr/sbin/newsyslog) input.type: log @timestamp: Jun 29, 2026 @ 15:07:00.057 location: 192.168.10.1 id: 1782727560.54787 timestamp: Jun 29, 2026 @ 15:07:00.057 _index: wazuh-archives-4.x-2026.06.29                                                                                                                      |
| > | Jun 29, 2026 @ 15:06:32.657 | predecoder.hostname: dhcpd[48472]: predecoder.timestamp: Jun 29 15:06:32 agent.name: wazuh-server agent.id: 000 manager.name: wazuh-server full_log: Jun 29 15:06:32 dhcpd[48472]: Sending Solicit input.type: log @timestamp: Jun 29, 2026 @ 15:06:32.657 location: 192.168.10.1 id: 1782727560.54787 timestamp: Jun 29, 2026 @ 15:06:32.657 _index: wazuh-archives-4.x-2026.06.29                                                                                                                                                         |
| > | Jun 29, 2026 @ 15:06:00.127 | predecoder.hostname: /usr/sbin/cron[76435]: predecoder.timestamp: Jun 29 15:06:00 agent.name: wazuh-server agent.id: 000 manager.name: wazuh-server full_log: Jun 29 15:06:00 /usr/sbin/cron[76435]: (root) CRD (/usr/sbin/newsyslog) input.type: log @timestamp: Jun 29, 2026 @ 15:06:00.127 location: 192.168.10.1 id: 1782727560.54787 timestamp: Jun 29, 2026 @ 15:06:00.127 _index: wazuh-archives-4.x-2026.06.29                                                                                                                      |
| > | Jun 29, 2026 @ 15:05:40.924 | predecoder.hostname: kali predecoder.program_name: systemd-timesyncd predecoder.timestamp: Jun 29 15:05:38 agent.ip: 192.168.10.100 agent.name: Kali_Linux agent.id: 001 manager.name: wazuh-server decoder.name: systemd full_log: Jun 29 15:05:38 kali systemd-timesyncd[450]: C ontacted time server 203.00.128.20:123 (1.debian.pool.ntp.org). input.type: log @timestamp: Jun 29, 2026 @ 15:05:40.924 location: journald id: 1782727560.54787 timestamp: Jun 29, 2026 @ 15:05:40.924 _index: wazuh-archives-4.x-2026.06.29             |
| > | Jun 29, 2026 @ 15:05:40.870 | predecoder.hostname: kali predecoder.program_name: systemd-timesyncd predecoder.timestamp: Jun 29 15:05:37 agent.ip: 192.168.10.100 agent.name: Kali_Linux agent.id: 001 manager.name: wazuh-server decoder.name: systemd full_log: Jun 29 15:05:37 kali systemd-timesyncd[450]: T imed out waiting for reply from 103.167.159.26:123 (1.debian.pool.ntp.org). input.type: log @timestamp: Jun 29, 2026 @ 15:05:40.870 location: journald id: 1782727560.54787 timestamp: Jun 29, 2026 @ 15:05:40.870 _index: wazuh-archives-4.x-2026.06.29 |
| > | Jun 29, 2026 @ 15:05:38.877 | predecoder.hostname: filterlog[37681]: predecoder.timestamp: Jun 29 15:05:38 agent.name: wazuh-server agent.id: 000 manager.name: wazuh-server full_log: Jun 29 15:05:38 filterlog[37681]: 09,,100000101,ent,match,pass,in,4,0xb6,,64,2077,0,CF,17,udp,76,192.168.10.100,103.55.68.158,44491,12 3,56 input.type: log @timestamp: Jun 29, 2026 @ 15:05:38.877 location: 192.168.10.1 id: 1782727536.54787 timestamp: Jun 29, 2026 @ 15:05:38.877 _index: wazuh-archives-4.x-2026.06.29                                                       |
| > | Jun 29, 2026 @ 15:05:38.877 | predecoder.hostname: filterlog[37681]: predecoder.timestamp: Jun 29 15:05:38 agent.name: wazuh-server agent.id: 000 manager.name: wazuh-server full_log: Jun 29 15:05:38 filterlog[37681]: 09,,100000101,ent,match,pass,in,4,0xb6,,64,15645,0,CF,17,udp,76,192.168.10.100,203.00.128.20,40771,12 3,56 input.type: log @timestamp: Jun 29, 2026 @ 15:05:38.877 location: 192.168.10.1 id: 1782727536.54787 timestamp: Jun 29, 2026 @ 15:05:38.877 _index: wazuh-archives-4.x-2026.06.29                                                      |
| > | Jun 29, 2026 @ 15:05:28.864 | predecoder.hostname: kali predecoder.program_name: systemd-timesyncd predecoder.timestamp: Jun 29 15:05:27 agent.ip: 192.168.10.100 agent.name: Kali_Linux agent.id: 001 manager.name: wazuh-server decoder.name: systemd full_log: Jun 29 15:05:27 kali systemd-timesyncd[450]: T imed out waiting for reply from 203.99.62.214:123 (1.debian.pool.ntp.org). input.type: log @timestamp: Jun 29, 2026 @ 15:05:28.864 location: journald id: 1782727526.54787 timestamp: Jun 29, 2026 @ 15:05:28.864 _index: wazuh-archives-4.x-2026.06.29  |
| > | Jun 29, 2026 @ 15:05:27.817 | predecoder.hostname: filterlog[37681]: predecoder.timestamp: Jun 29 15:05:27 agent.name: wazuh-server agent.id: 000 manager.name: wazuh-server full_log: Jun 29 15:05:27 filterlog[37681]: 09,,100000101,ent,match,pass,in,4,0xb6,,64,63207,0,CF,17,udp,76,192.168.10.100,103.167.159.26,34217,12 3,56 input.type: log @timestamp: Jun 29, 2026 @ 15:05:27.817 location: 192.168.10.1 id: 1782727527.54787 timestamp: Jun 29, 2026 @ 15:05:27.817 _index: wazuh-archives-4.x-2026.06.29                                                     |
| > | Jun 29, 2026 @ 15:05:17.717 | predecoder.hostname: filterlog[37681]: predecoder.timestamp: Jun 29 15:05:17 agent.name: wazuh-server agent.id: 000 manager.name: wazuh-server full_log: Jun 29 15:05:17 filterlog[37681]: 09,,100000101,ent,match,pass,in,4,0xb6,,64,36258,0,CF,17,udp,76,192.168.10.100,203.99.62.214,40324,12 3,56 input.type: log @timestamp: Jun 29, 2026 @ 15:05:17.717 location: 192.168.10.1 id: 1782727517.54787 timestamp: Jun 29, 2026 @ 15:05:17.717 _index: wazuh-archives-4.x-2026.06.29                                                      |
| > | Jun 29, 2026 @ 15:05:16.717 | predecoder.hostname: filterlog[37681]: predecoder.timestamp: Jun 29 15:05:16 agent.name: wazuh-server agent.id: 000 manager.name: wazuh-server full_log: Jun 29 15:05:16 filterlog[37681]: 09,,100000101,ent,match,pass,in,4,0xb6,,64,1043,0,CF,17,udp,76,192.168.10.100,103.55.68.158,32030,12 3,56 input.type: log @timestamp: Jun 29, 2026 @ 15:05:16.717 location: 192.168.10.1 id: 1782727516.54787 timestamp: Jun 29, 2026 @ 15:05:16.717 _index: wazuh-archives-4.x-2026.06.29                                                       |

## Summary

In this project, a virtual security lab was created using Oracle VirtualBox with pfSense as the firewall and Wazuh as the Security Information and Event Management (SIEM) platform. pfSense was configured to filter network traffic, restrict administrative access, and block traffic from selected countries using pfBlockerNG. Firewall logs were then forwarded to the Wazuh server through Syslog, where they were successfully collected, monitored, and analyzed in the Wazuh Dashboard.

The project demonstrates how pfSense and Wazuh can be integrated to provide centralized log monitoring, network visibility, and improved security monitoring in a small enterprise environment.